Solution

With usual notations in triangle ABC, we have $\overrightarrow{BC} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, and $\overrightarrow{AB} = \vec{c}$. Then

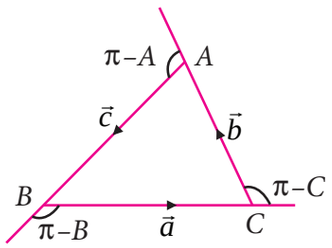


Fig. 6.7

$$|\overrightarrow{BC}| = a,\ |\overrightarrow{CA}| = b,\ |\overrightarrow{AB}| = c \text{ and } \overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = \vec{0}$$

So, $\overrightarrow{BC} = -\overrightarrow{CA} - \overrightarrow{AB}$

Applying dot product, we get

$$\overrightarrow{BC} \cdot \overrightarrow{BC} = -\overrightarrow{BC} \cdot \overrightarrow{CA} - \overrightarrow{BC} \cdot \overrightarrow{AB}$$

$$\Rightarrow |\overrightarrow{BC}|^2 = -|\overrightarrow{BC}||\overrightarrow{CA}| \cos(\pi - C) - |\overrightarrow{BC}||\overrightarrow{AB}| \cos(\pi - B)$$

$$\Rightarrow a^2 = ab\cos C + ac\cos B$$

Therefore $a = b\cos C + c\cos B$. The results in (ii) and (iii) are proved in a similar way. ■

Example 6.3

By vector method, prove that $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$.

Solution

Let $\hat{a} = \overrightarrow{OA}$ and $\hat{b} = \overrightarrow{OB}$ be the unit vectors and which make angles α and β, respectively, with positive x-axis, where A and B are as in the Fig. 6.8. Draw AL and BM perpendicular to the x-axis. Then $|\overrightarrow{OL}| = |\overrightarrow{OA}|\cos\alpha = \cos\alpha$, $|\overrightarrow{LA}| = |\overrightarrow{OA}|\sin\alpha = \sin\alpha$.

So, $\overrightarrow{OL} = |\overrightarrow{OL}|\,\hat{i} = \cos\alpha\hat{i}$, $\overrightarrow{LA} = \sin\alpha(-\hat{j})$.

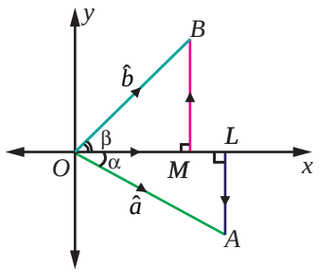


Fig. 6.8

Therefore, $\hat{a} = \overrightarrow{OA} = \overrightarrow{OL} + \overrightarrow{LA} = \cos\alpha\hat{i} - \sin\alpha\hat{j}$. ... (1)

Similarly, $\hat{b} = \cos\beta\hat{i} + \sin\beta\hat{j}$ ... (2)

The angle between $\hat{a}$ and $\hat{b}$ is $\alpha + \beta$ and so,

$$\hat{a} \cdot \hat{b} = |\hat{a}|\,|\hat{b}|\cos(\alpha + \beta) = \cos(\alpha + \beta) \quad \text{... (3)}$$

On the other hand, from (1) and (2)

$$\hat{a} \cdot \hat{b} = (\cos\alpha\hat{i} - \sin\alpha\hat{j}) \cdot (\cos\beta\hat{i} + \sin\beta\hat{j}) = \cos\alpha\cos\beta - \sin\alpha\sin\beta. \quad \text{... (4)}$$

From (3) and (4), we get $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$. ■

Example 6.4

With usual notations, in any triangle ABC, prove by vector method that $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.

Solution

With usual notations in triangle ABC, we have $\overrightarrow{BC} = \vec{a}, \overrightarrow{CA} = \vec{b}$, and $\overrightarrow{AB} = \vec{c}$. Then $|\overrightarrow{BC}| = a$, $|\overrightarrow{CA}| = b$, and $|\overrightarrow{AB}| = c$.

Since in ΔABC, $\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = 0$, we have $\overrightarrow{BC} \times (\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB}) = \vec{0}$.

Simplification gives,

$$\overrightarrow{BC} \times \overrightarrow{CA} = \overrightarrow{AB} \times \overrightarrow{BC}. \quad \text{... (1)}$$

Similarly, since $\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB} = \vec{0}$, we have

$$\overrightarrow{CA} \times (\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB}) = \vec{0}.$$

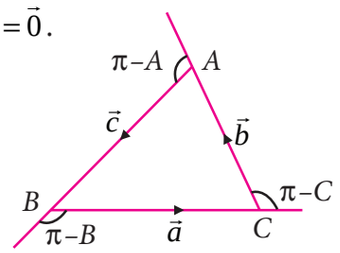


Fig. 6.9